

Challenge (Habanero)

- Q1.** A server's latency threshold is defined by the logic $|L - 15| \leq 5$. Which of the following pings (L) would trigger a 'Connection Timed Out' error?
- (A) 10
(B) 12
(C) 15
(D) 19
(E) 21
- Q2.** A cloud storage monitor tracks the deviation from a daily quota using $|u - 500|$ GB. If the current usage u is 425 GB, what is the absolute deviation?
- (A) -75
(B) 0
(C) 75
(D) 125
(E) 500
- Q3.** In a combat game, your 'Fury Meter' is calculated as $2 \cdot |s| - |-10|$, where s is your negative score. If $s = -42$, evaluate the meter level.
- (A) 74
(B) 84
(C) 94
(D) 32
(E) 52
- Q4.** A script executes only if $|x| > |y|$. If $x = -18$ and $y = 20$, which modification to x makes the condition true?
- (A) Add 1 to x
(B) Subtract 3 from x
(C) Add 2 to x
(D) Multiply x by 0
(E) Divide x by -2
- Q5.** A sprite moves from coordinate $x = -14$ to $x = 22$ on a horizontal grid. What is the absolute distance traveled by the sprite?
- (A) 8
(B) 14
(C) 22
(D) 36
(E) 40
- Q6.** Evaluate the signal noise expression used in a data packet header: $|| - 12| - |-15|| \cdot |-3|$.
- (A) -9
(B) 3
(C) 9
(D) 27
(E) 81
- Q7.** A variable k represents a packet loss offset. For any integer value of k , which of the following is always true about $|-k|$?
- (A) $|-k| = -k$
(B) $|-k| = k$
(C) $|-k| \geq 0$
(D) $|-k| < 0$
(E) $|-k| = 0$
- Q8.** A leaderboard rank change is $d = |a - b|$. If a player's previous rank index was $a = -105$ and current is $b = -85$, find d .
- (A) -190
(B) -20
(C) 20
(D) 190
(E) -210

Open Ended Questions (FRQ)

- Q9.** A streaming buffer size B is calculated as $B = |t_1 - t_2| + |t_2 - t_3|$. If timestamps are $t_1 = -15\text{ms}$, $t_2 = 10\text{ms}$, and $t_3 = -5\text{ms}$, calculate the total value of B .
- Q10.** In a physics engine, the distance between two nodes on the x-axis is $|x_1 - x_2|$. If $x_1 = -45$ and the absolute distance is exactly 100 units, find all possible integer values for x_2 .

Chef's Tips

- Remind students that absolute value measures magnitude, never returning a negative result.
- Encourage drawing a 1D coordinate line for distance-based problems.
- Watch for signs when substituting negative values into expressions like $|-k|$.